ΟΙΚΟΝΟΜΙΚΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΑΘΗΝΩΝ
ATHENS UNIVERSITY OF ECONOMICS AND BUSINESS
ΣΧΟΛΗ ΔΙΟΙΚΗΣΗΣ ΕΠΙΧΕΙΡΗΣΕΩΝ
SCHOOL OF BUSINESS
ΤΜΗΜΑ ΔΙΟΙΚΗΤΙΚΗΣ ΕΠΙΣΤΗΜΗΣ & ΤΕΧΝΟΛΟΓΙΑΣ
DEPARTMENT OF MANAGEMENT SCIENCE & TECHNOLOGY

Επιχειρηματική Ευφυΐα & Ανάλυση Μεγάλων Δεδομένων

Global Terrorism Attacks

Meet the Team

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Agenda

- Data Overview
- Data Transformation
- Data Warehouse
- Data Visualization
- Data Mining

✓Data Overview

Global Terrorism Database

More than 180,000 terrorist attacks worldwide, 1970-2017

- 155 MB
- 135 attributes
- 1,100 voters on Kaggle
- Maintained by researchers at the National Consortium for the Study of Terrorism and Responses to Terrorism (START), University of Maryland

✓Data Transformation

Final Dataset

- **20** attributes included
- **104,000** records (60%) included
- Attack information: **date**, **location**, **target**, **attacker**, **success**, **weapon**, **suicide**, **victims**
- Added attribute of **frequency** for attacks of same city but different town
- Kept **null** values of attacker group and victims
- Binary Values: 1, 0 → **True**, **False**

✓Record Example

day:
10

region:
Western Europe

extended:
False

city:
Athens

month:
1

year:
2017

country:
Greece

suicide:
False

fatalities:
0

province:
Attica

attackID:
1099154

frequency:
1

wounded:
1

target_type:
Police

nationality:
Greece

weapon:
Firearms

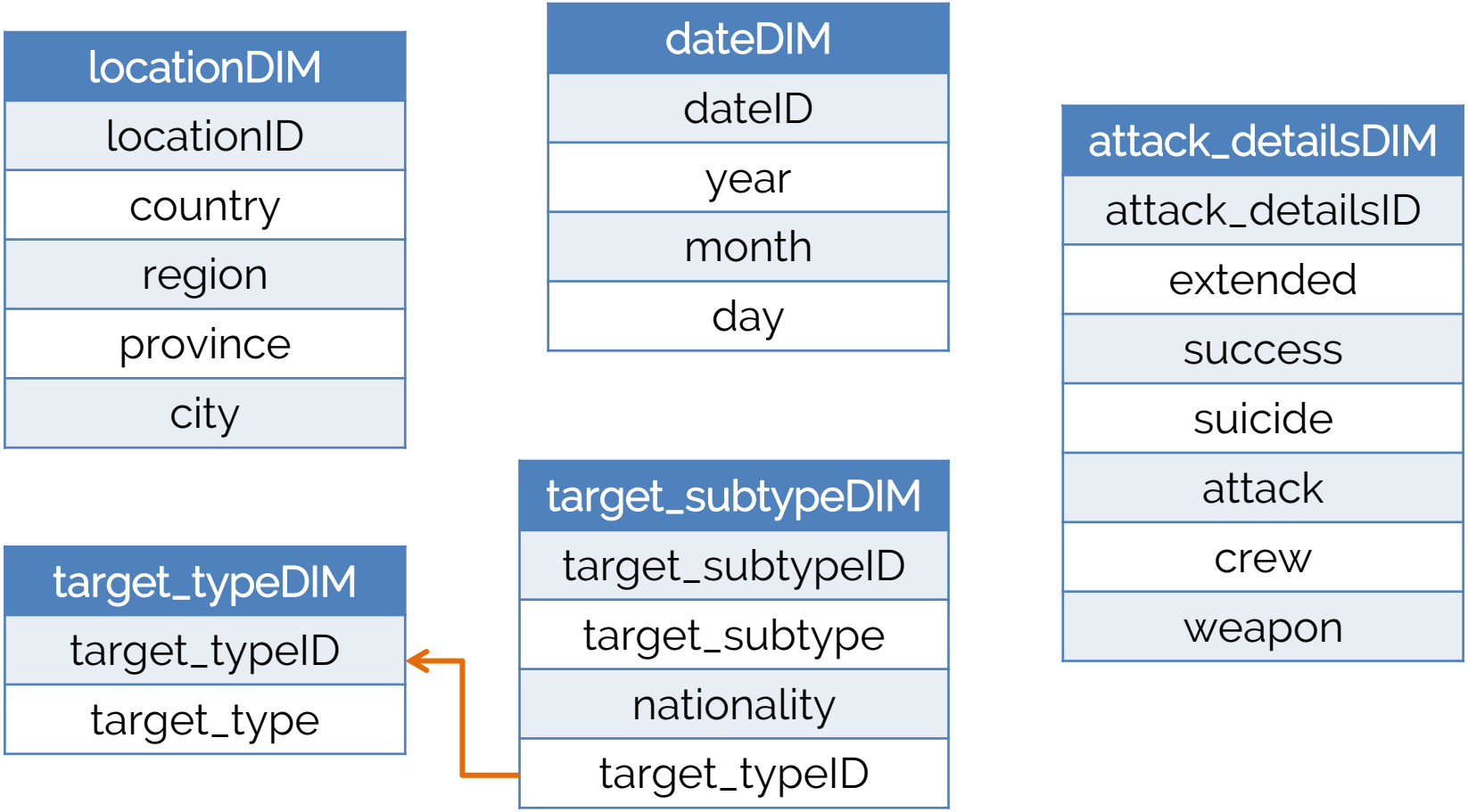
success:
True

target_subtype:
Police Patrol (including vehicles and convoys)

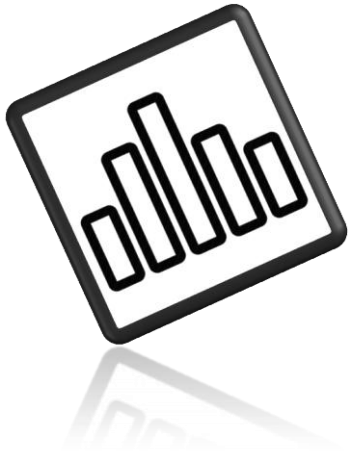
crew:
Organization for Revolutionary Self Defense

attack:
Armed Assault

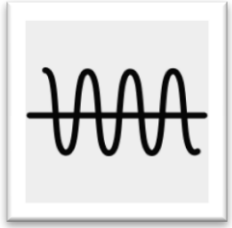
✓Dimensions


locationDIM
locationID
country
region
province
city
dateDIM
dateID
year
month
day
attack_detailsDIM
attack_detailsID
extended
success
suicide
attack
crew
weapon
target_subtypeDIM
target_subtypeID
target_subtype
nationality
target_typeID
target_typeDIM
target_typeID
target_type

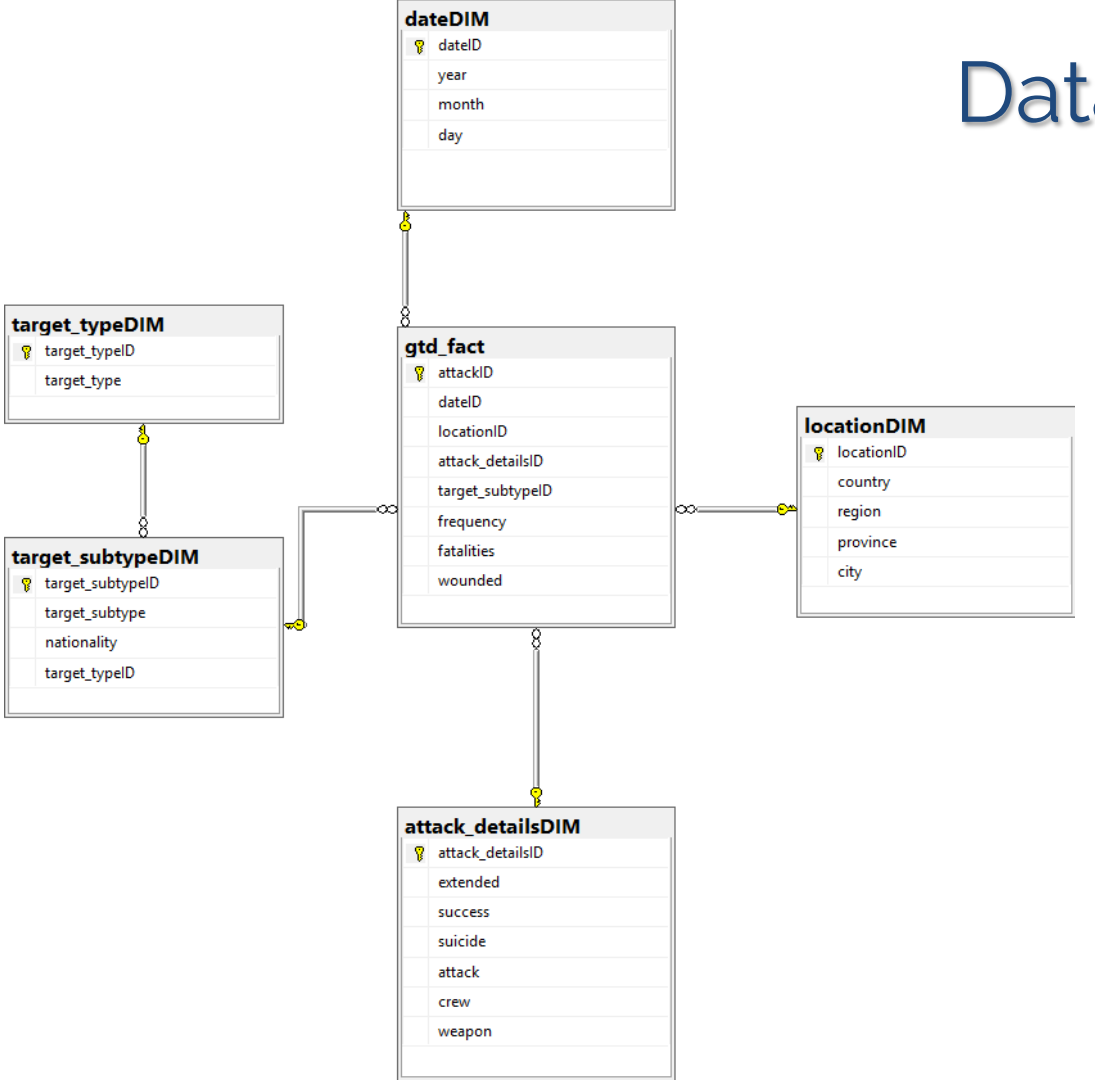
✓Measures



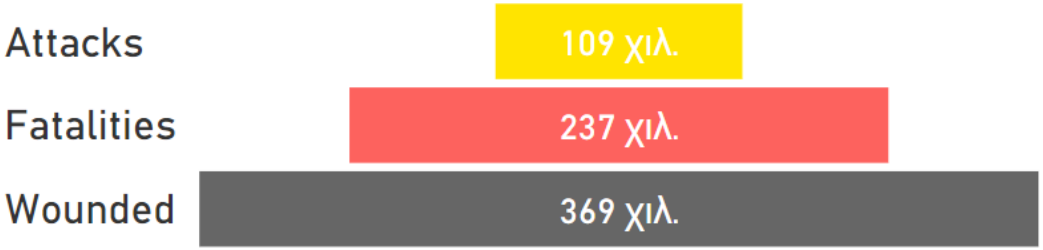
- frequency
- fatalities
- wounded

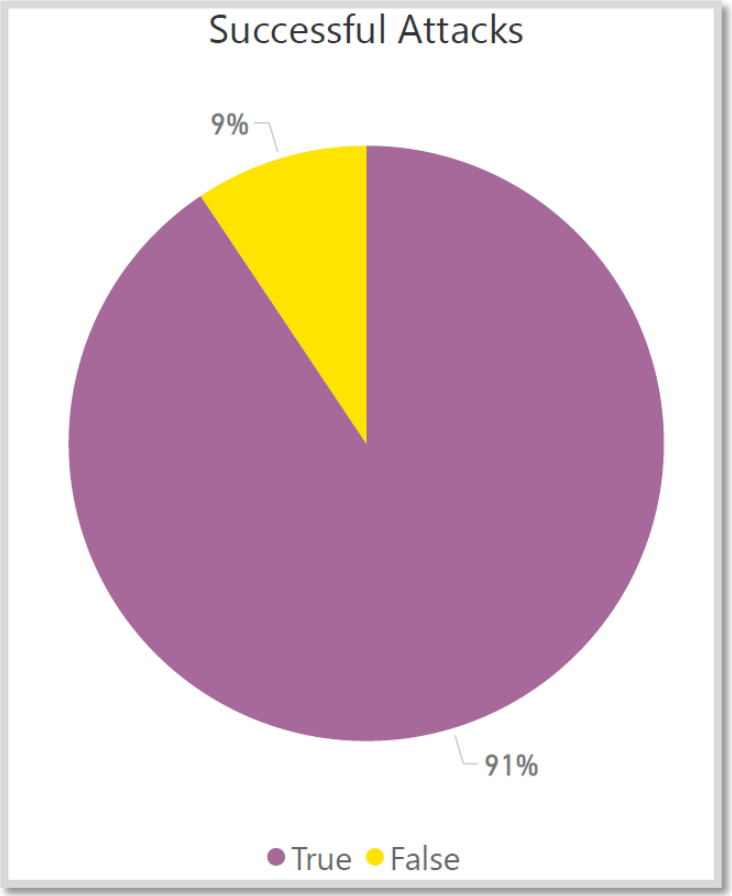


Database Diagram

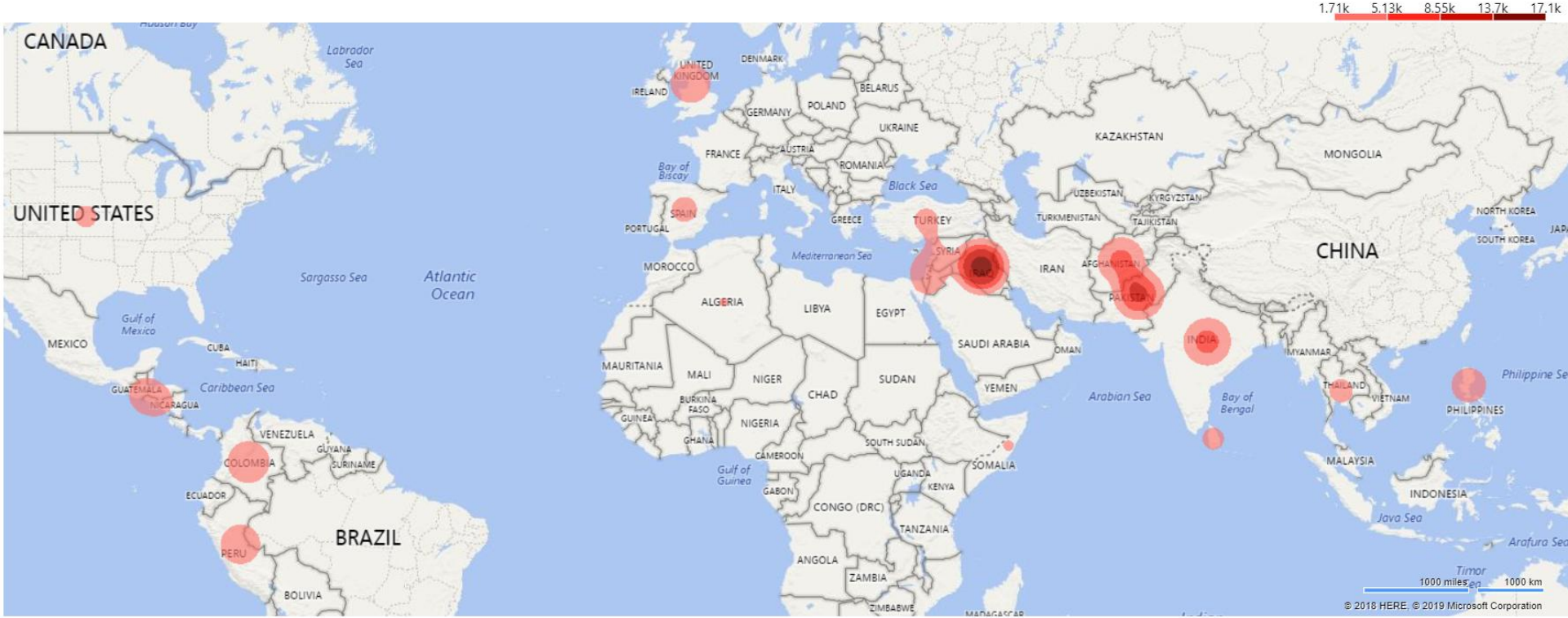


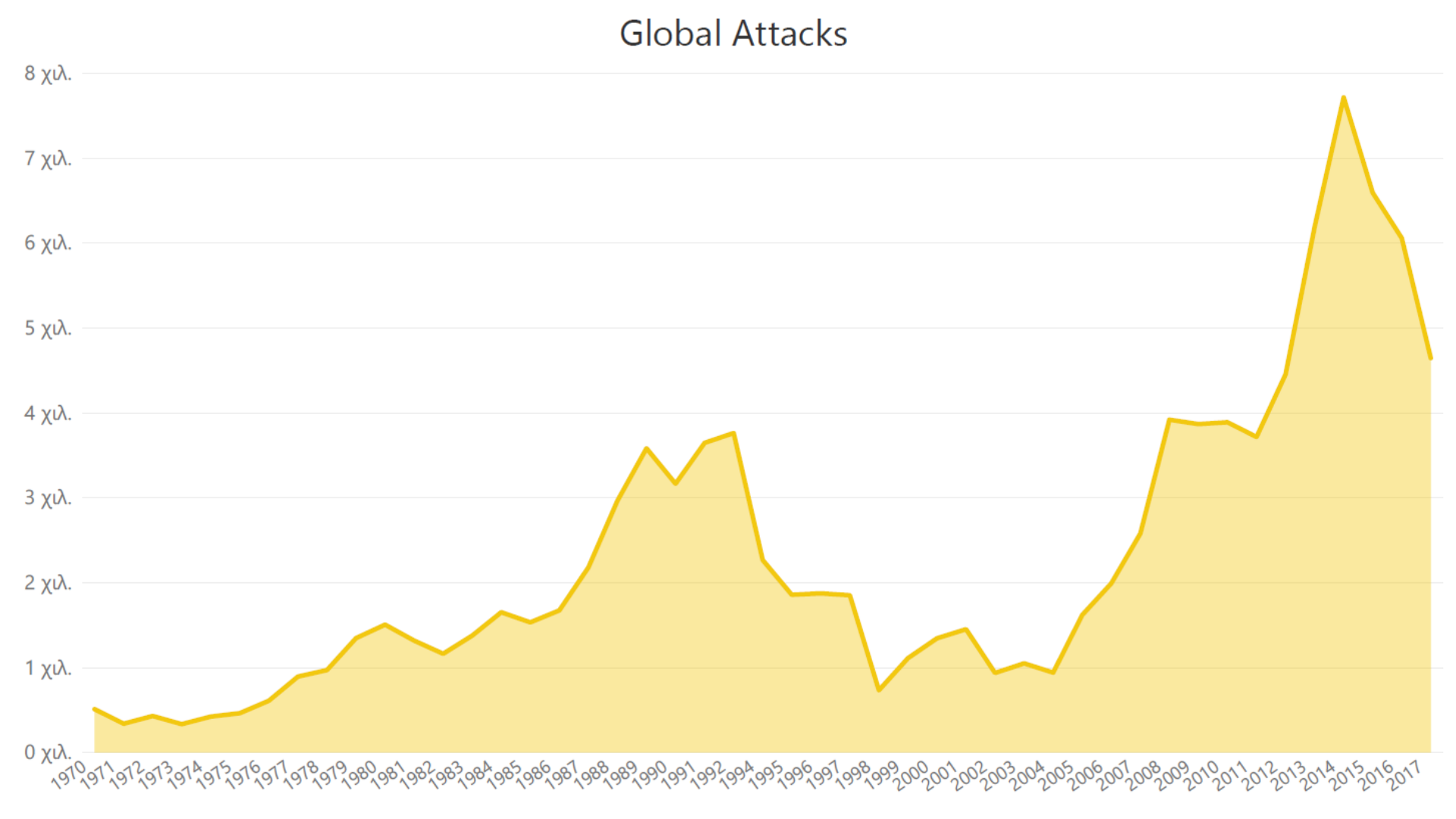
✓Data Visualization


Attacks
109 χιλ.
Fatalities
237 χιλ.
Wounded
369 χιλ.


Successful Attacks
9%
91%
True
False

Global Attacks




Global Attacks
8 χιλ.
7 χιλ.
6 χιλ.
5 χιλ.
4 χιλ.
3 χιλ.
2 χιλ.
1 χιλ.
0 χιλ.
1970
1971
1972
1973
1974
1975
1976
1977
1978
1979
1980
1981
1982
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1984
1985
1986
1987
1988
1989
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1998
1999
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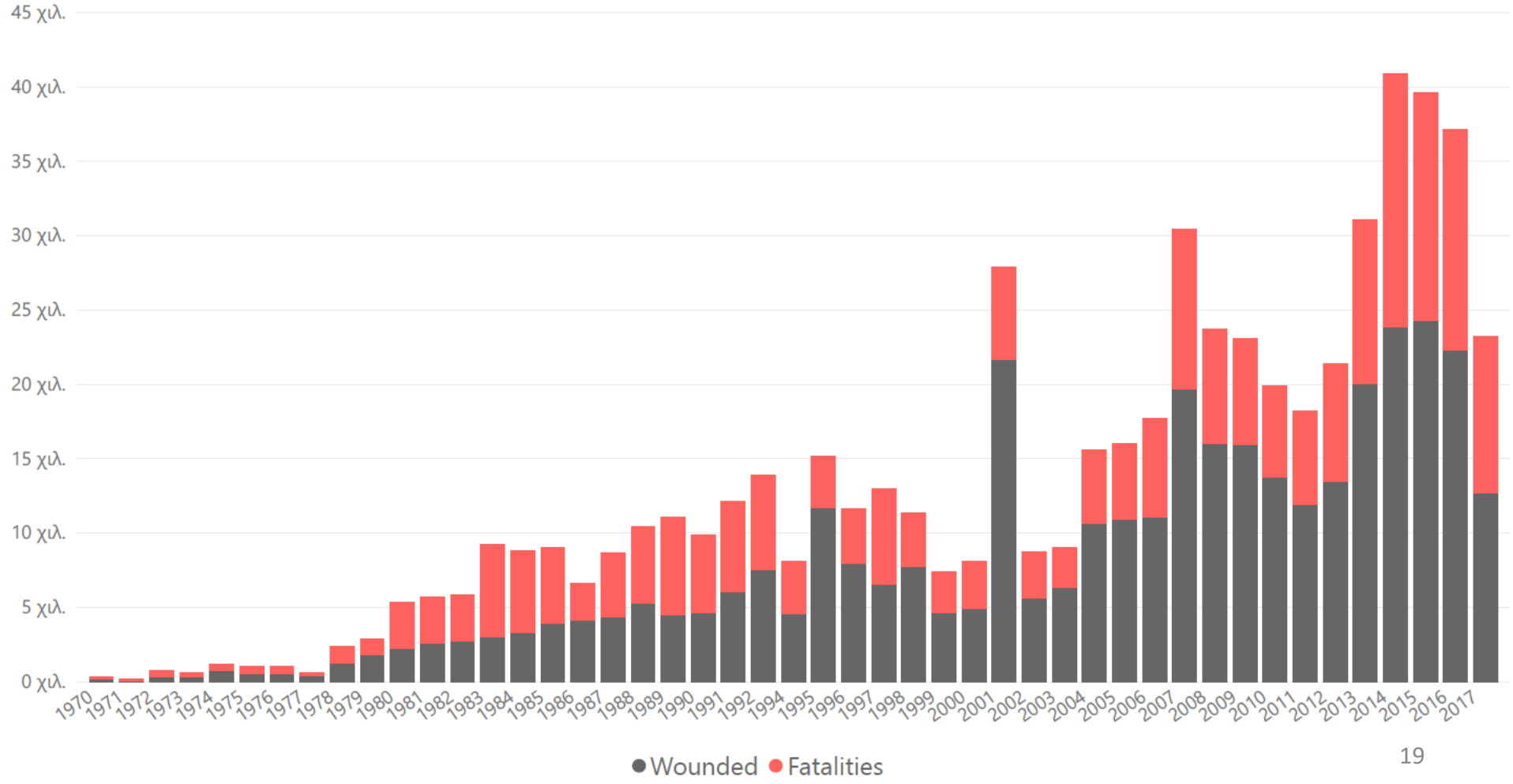
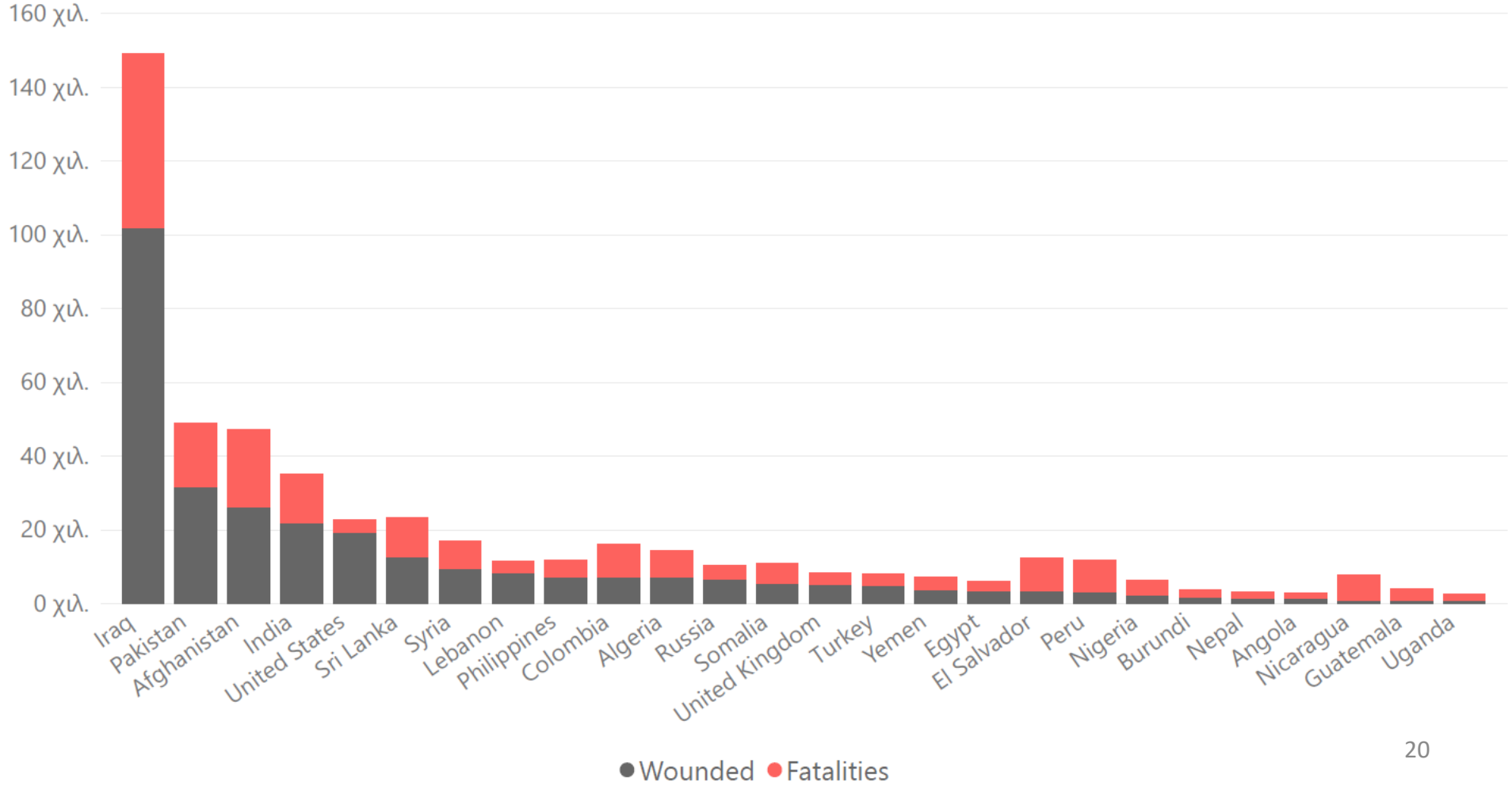
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017

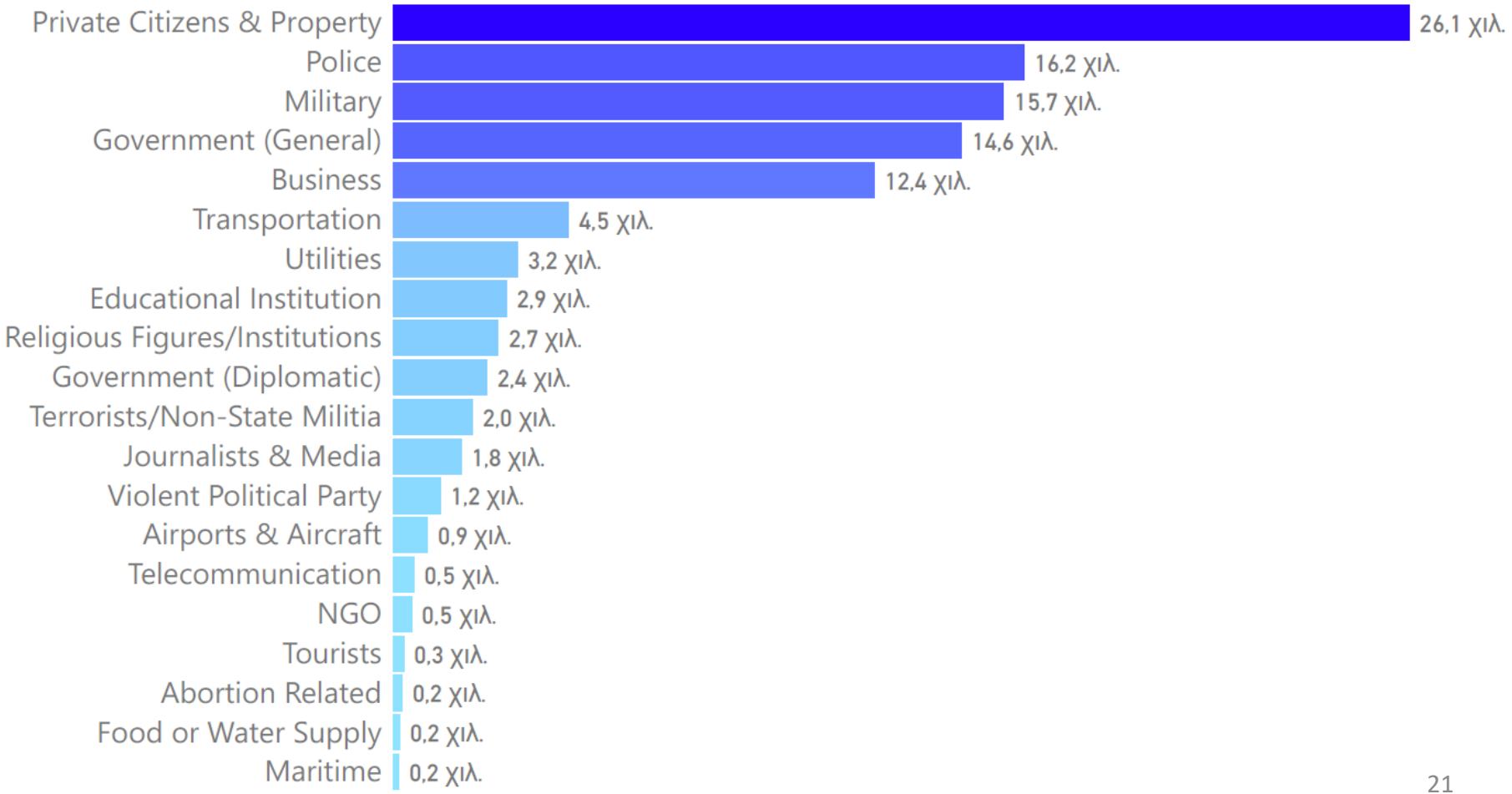
Wounded-Fatalities



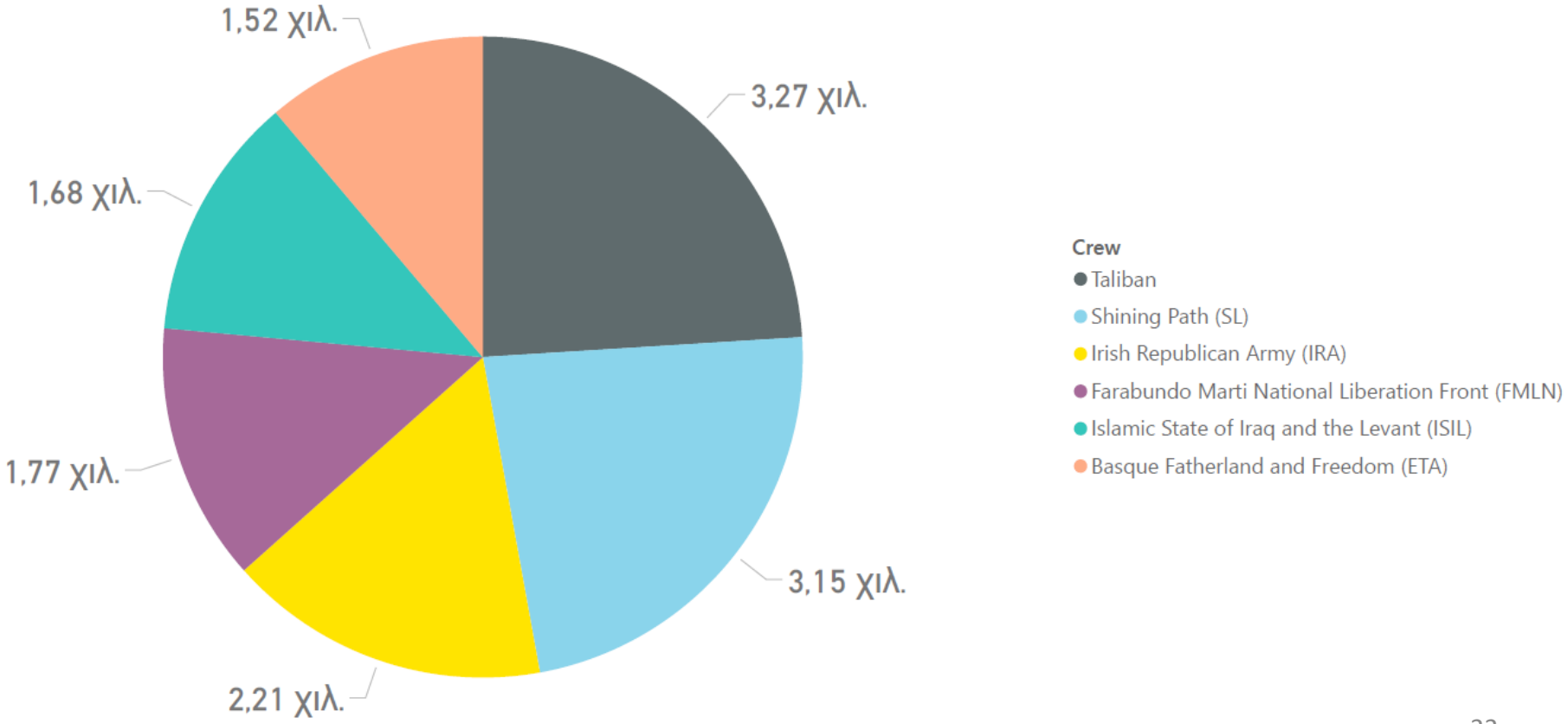
Wounded-Fatalities per Country



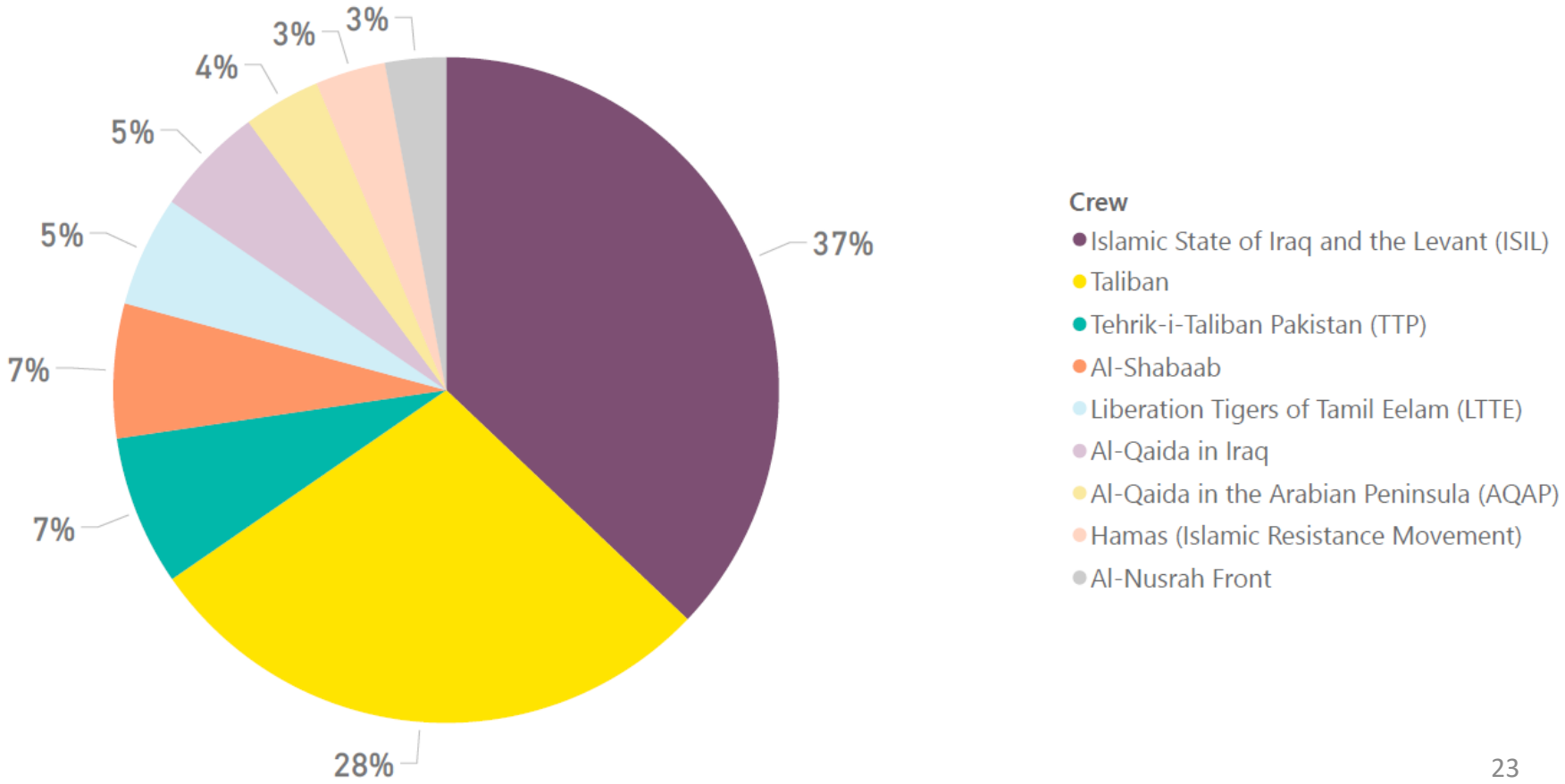
Attacks per Target

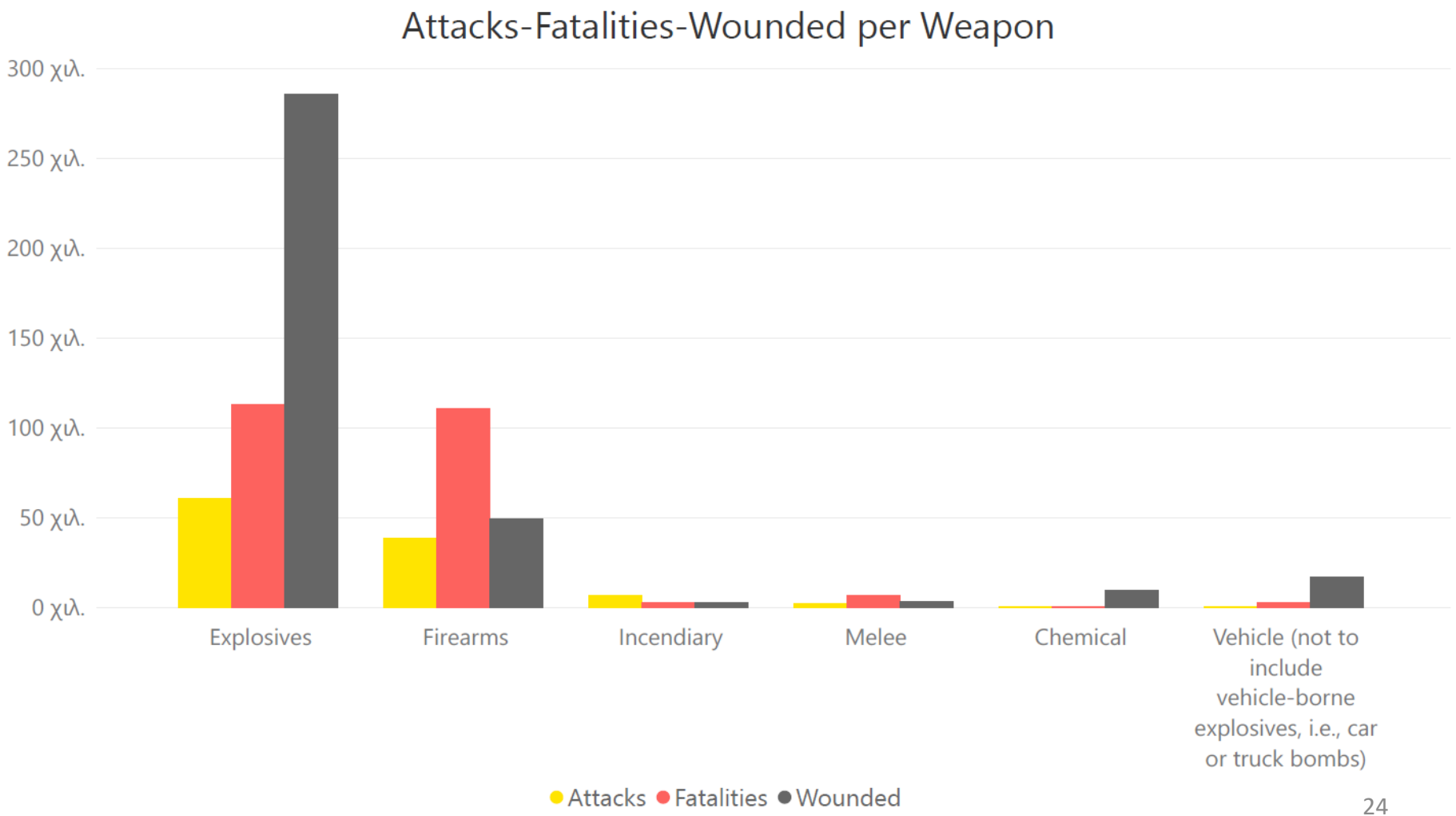


Top Crews' Attacks

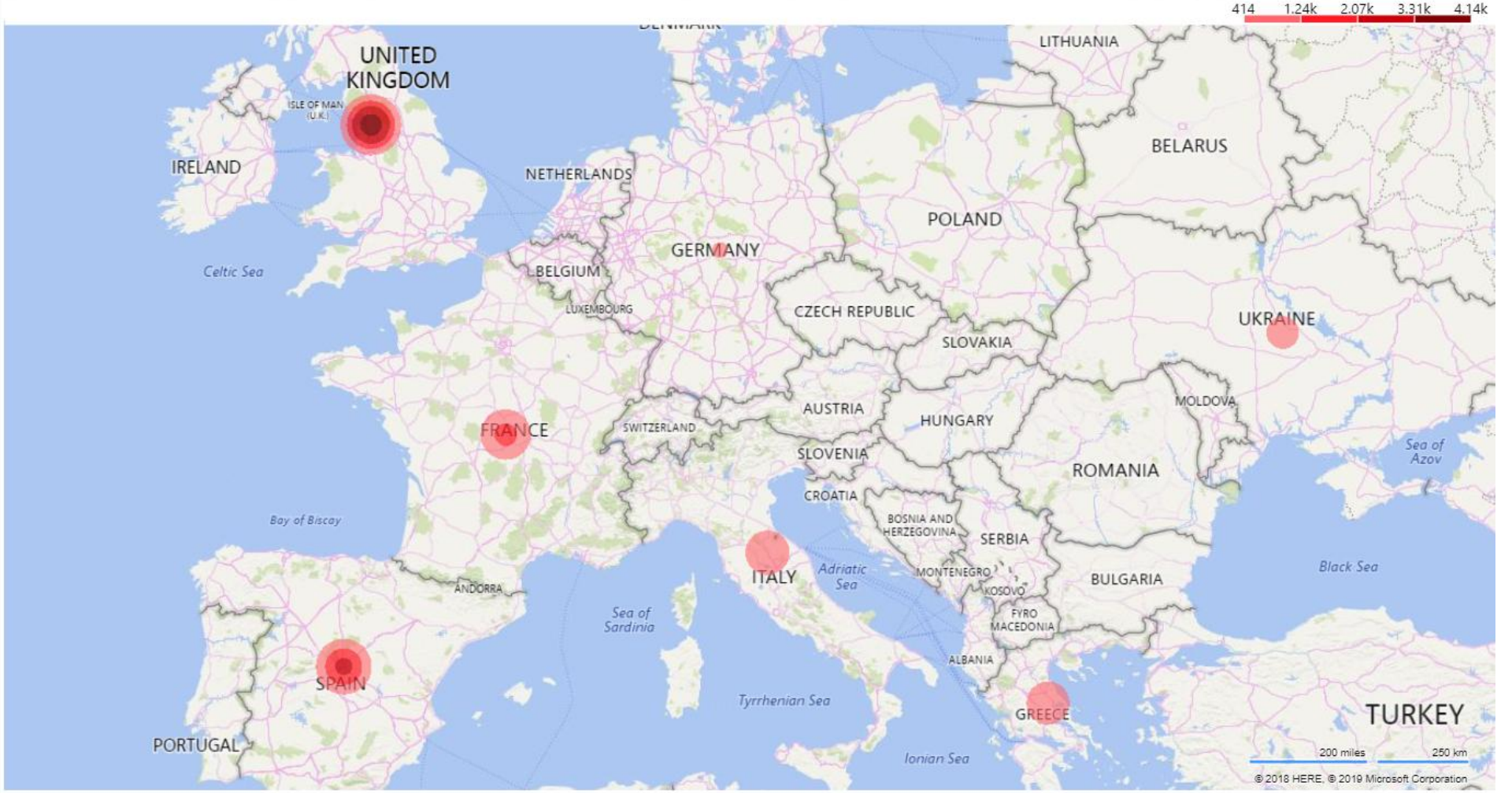


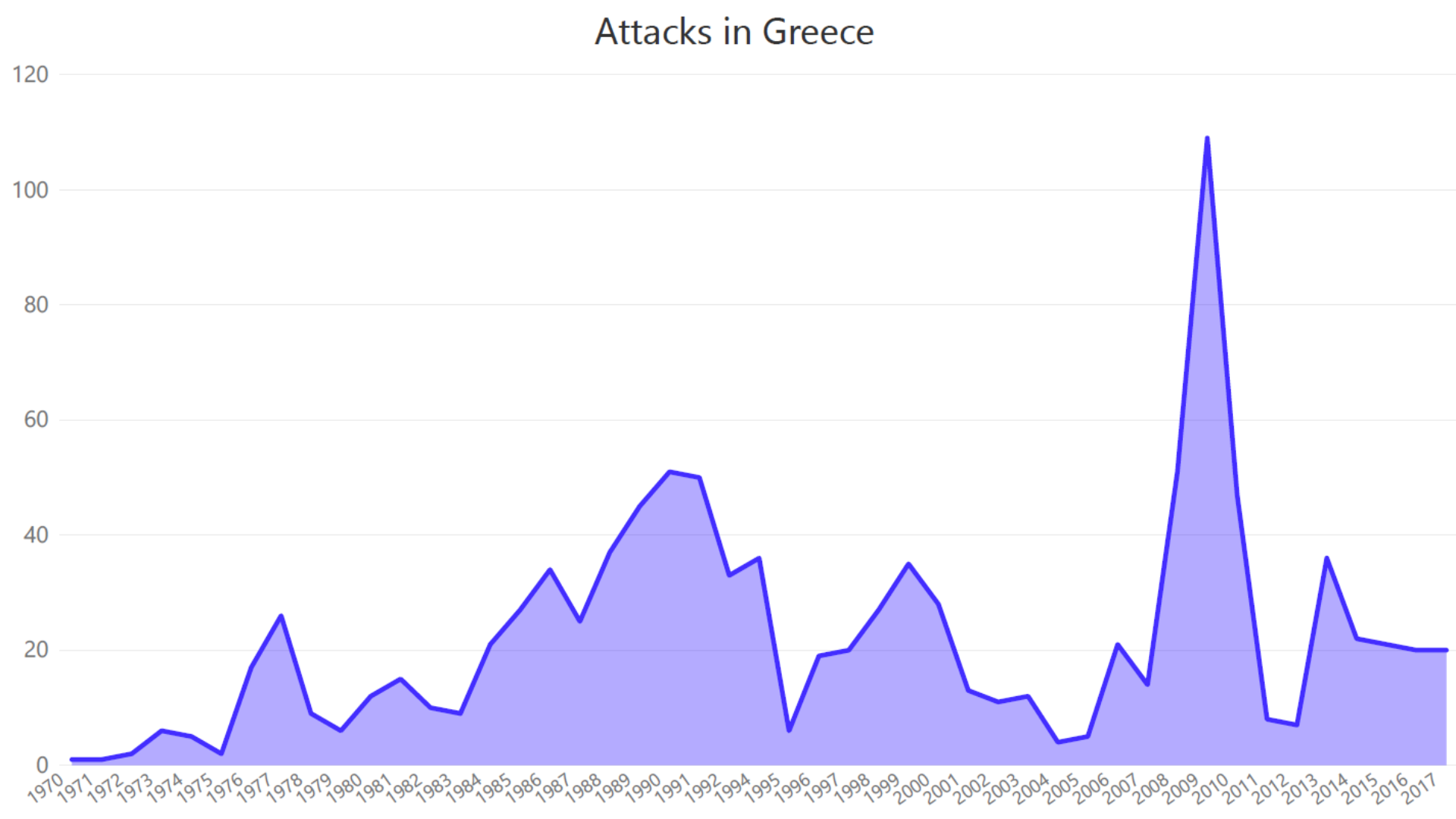
Suicidal Crews




Attacks-Fatalities-Wounded per Weapon
300 χιλ.
250 χιλ.
200 χιλ.
150 χιλ.
100 χιλ.
50 χιλ.
0 χιλ.
Explosives
Firearms
Incendiary
Melee
Chemical
Vehicle (not to include vehicle-borne explosives, i.e., car or truck bombs)
Attacks
Fatalities
Wounded

Attacks in Europe




Attacks in Greece
120
100
80
60
40
20
0
1970
1971
1972
1973
1974
1975
1976
1977
1978
1979
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
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1998
1999
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2013
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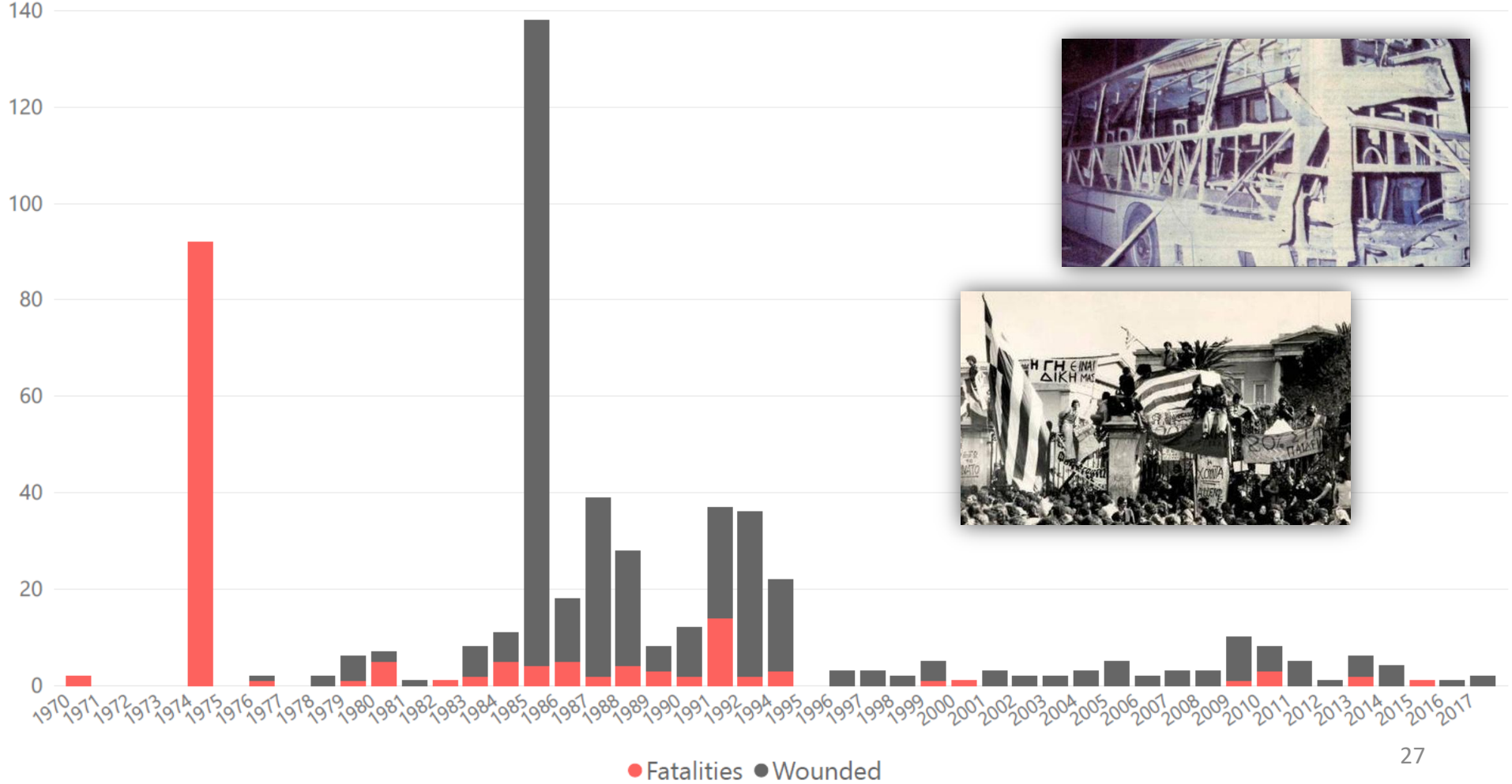
2015
2016
2017

Fatalities-Wounded in Greece



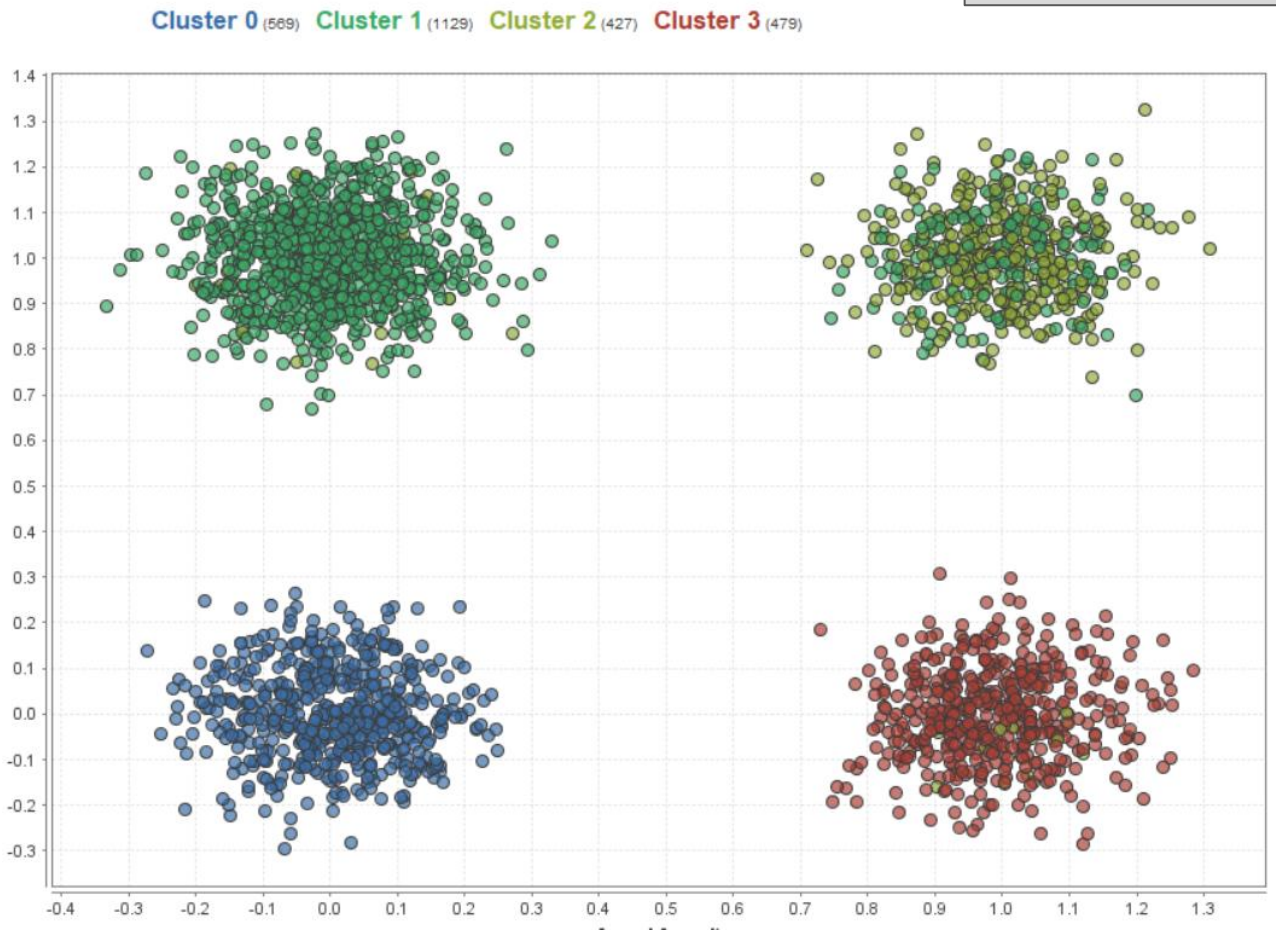
✓Data Mining

1. Clustering Crews based on Attacks

DBI: 1.17
Avg. Cluster Distance: 0.5

Bombers (1,130)

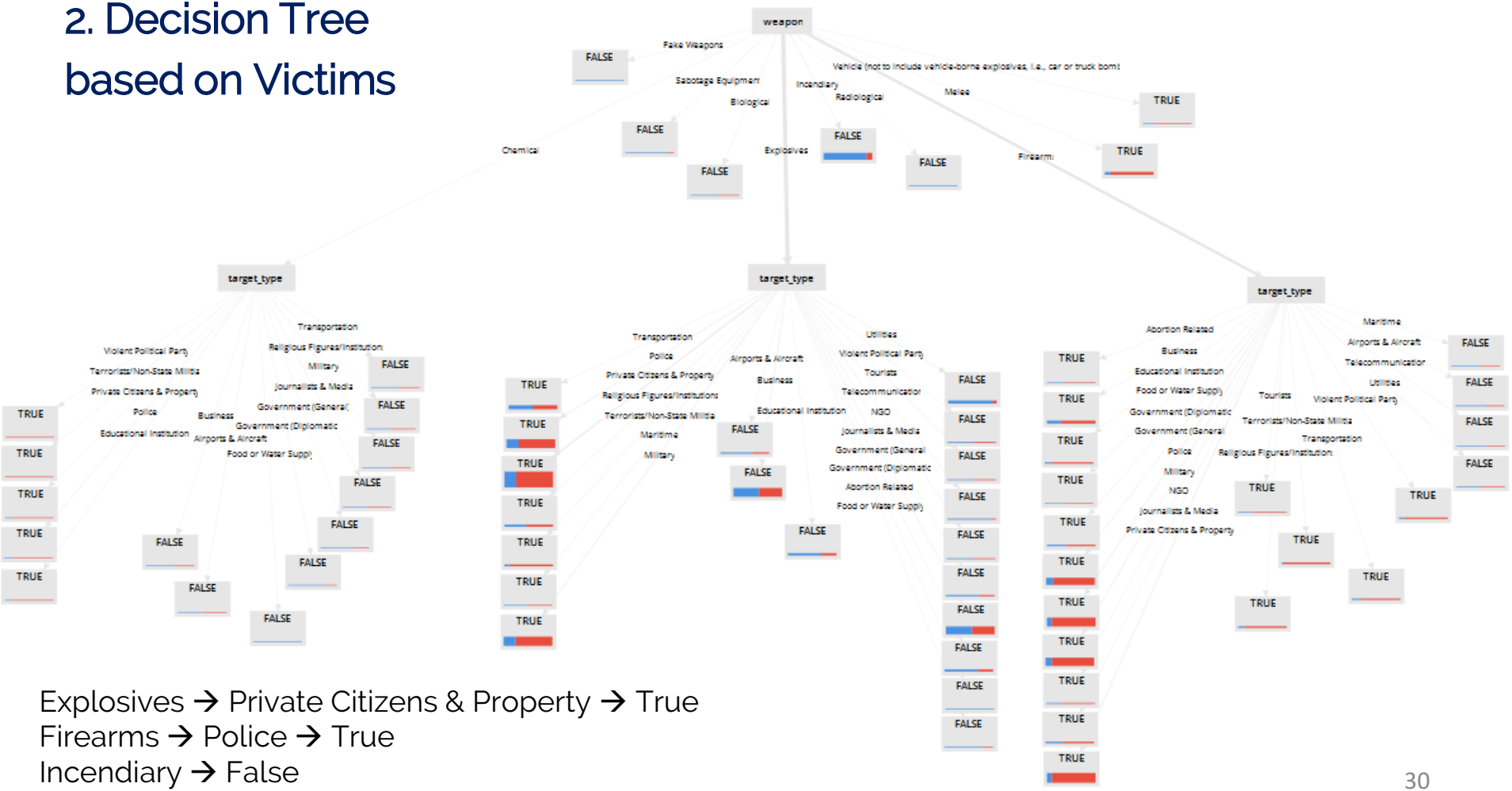
Unarmed Demolishers (570)



Hijackers-Kidnappers (430)

Gunners (480)

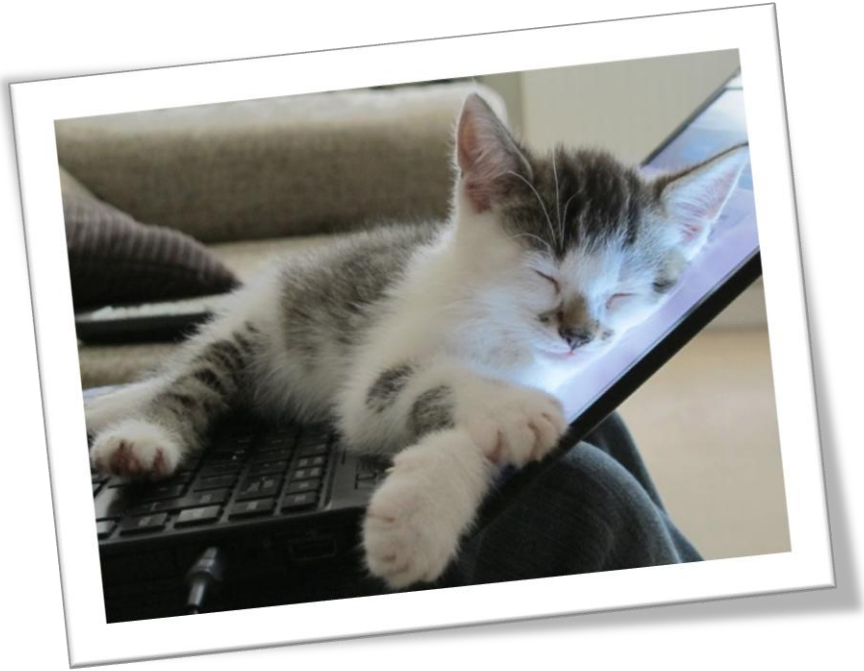
2. Decision Tree based on Victims


weapon
Fake Weapons
FALSE
Sabotage Equipment
FALSE
Biological
FALSE
Chemical
Explosives
FALSE
Incendiary
Radiological
FALSE
Vehicle (not to include vehicle-borne explosives, i.e., car or truck bomb)
Melee
FALSE
Firearm
TRUE
TRUE
target_type
Violent Political Party
Terrorists/Non-State Militia
Private Citizens & Property
Police
Educational Institution
Business
Airports & Aircraft
Food or Water Supply
Government (Diplomatic
Government (General
Journalists & Media
Military
Religious Figures/Institution
Transportation
TRUE
TRUE
TRUE
TRUE
TRUE
TRUE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
target_type
Transportation
Police
Private Citizens & Property
Religious Figures/Institution
Terrorists/Non-State Militia
Maritime
Military
Airports & Aircraft
Business
Educational Institution
Utilities
Violent Political Party
Tourists
Telecommunication
NGO
Journalists & Media
Government (General
Government (Diplomatic
Abortion Related
Food or Water Supply
TRUE
TRUE
TRUE
TRUE
TRUE
TRUE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
FALSE
target_type
Abortion Related
Business
Educational Institution
Food or Water Supply
Government (Diplomatic
Government (General
Police
Military
NGO
Journalists & Media
Private Citizens & Property
Tourists
Terrorists/Non-State Militia
Religious Figures/Institution
Violent Political Party
Transportation
Maritime
Airports & Aircraft
Telecommunication
Utilities
TRUE
TRUE
TRUE
TRUE
TRUE
TRUE

2. Decision Tree based on Victims (2)

	true FALSE	true TRUE	class precision
pred. FALSE	5,653	2,938	65.80%
pred. TRUE	4,430	18,061	80.30%
class recall	56.06%	86.01%	

3. Association Rules on Targets

		Confidence	Support	Lift
Military	Private Citizens & Property	56.9%	12.7%	1.549
Police	Private Citizens & Property	56.8%	13.6%	1.548
Police	Government (General)	53.9%	12.9%	1.618
Business	Private Citizens & Property	49.5%	15.2%	1.348

Thank you!